

CIS 6270 Reference Formulas

Section numbers refer to the Fall 2026 lecture notes.

Probability and Gaussians

Mean and covariance of an affine map §1.14.6

$$\mathbb{E}[AX + b] = A\mu + b, \quad \text{Cov} = A\Sigma A^\top$$

Gaussian noising, independent terms §1.8

$$X_t = \alpha_t X_0 + \sigma_t \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

$$\text{Cov}(X_t) = \alpha_t^2 \text{Cov}(X_0) + \sigma_t^2 I$$

Score of a Gaussian §1.31

$$\nabla_x \log \mathcal{N}(x; m, \sigma^2 I) = -\frac{x - m}{\sigma^2}$$

Bayes' rule §1.10

$$p(x | y) = \frac{p(y | x) p(x)}{p(y)}$$

Total probability and posterior weight §1.10

$$p(x) = \int p(x | y) p(y) dy, \quad w(y | x) = \frac{p(x | y) p(y)}{p(x)}$$

Linear algebra

Diagonalization, matrix exponential and its determinant §1.16.3 §1.18.2

$$A = V\Lambda V^{-1}, \quad e^{tA} = V e^{t\Lambda} V^{-1}$$

$$\det e^{hM} = \prod_i e^{hm_i} = e^{h \text{tr} M}$$

Divergence is a Jacobian trace §1.29

$$\nabla \cdot v = \text{tr} J_v = \sum_i \partial_i v_i$$

Laplacian §1.32

$$\nabla^2 f = \nabla \cdot \nabla f = \sum_i \partial_i^2 f$$

Change of variables

Pushforward §1.20

$$\mathbb{P}(Y \in B) = \mathbb{P}(X \in \phi^{-1}(B))$$

Density under an invertible map §1.24

$$p_Y(\phi(x)) = \frac{p_X(x)}{|\det J_\phi(x)|}$$

Log-density form §1.25

$$\log p_Y(\phi(x)) = \log p_X(x) - \log |\det J_\phi(x)|$$

Flows and densities

Trajectory and flow map §2.2

$$\frac{dx_t}{dt} = v_t(x_t), \quad x_t = \phi_t(x_0)$$

Euler step §2.5

$$\hat{x}_{k+1} = \hat{x}_k + \Delta t v_{t_k}(\hat{x}_k)$$

Continuity equation and flux §2.7

$$\partial_t p_t + \nabla \cdot (p_t v_t) = 0, \quad j_t = p_t v_t$$

Score identity §3.32

$$s_t = \nabla \log p_t \iff p_t s_t = \nabla p_t$$

Log density along a trajectory §2.8

$$\frac{d}{dt} \log p_t(x_t) = -\nabla \cdot v_t(x_t)$$

$$\log p_1(x_1) = \log p_0(x_0) - \int_0^1 \nabla \cdot v_t dt$$

Flow matching

Linear interpolant and target §2.12

$$X_t = (1-t)X_0 + tX_1, \quad U = X_1 - X_0$$

General schedule §2.18

$$X_t = \alpha_t X_0 + \beta_t X_1, \quad U = \dot{\alpha}_t X_0 + \dot{\beta}_t X_1$$

Marginal velocity §2.13

$$v_t(x) = \mathbb{E}[U_t | X_t = x]$$

Least squares gives the conditional mean §2.14.1

$$\arg \min_a \mathbb{E}[\|a - U\|^2 | X_t = x] = \mathbb{E}[U | X_t = x]$$

Conditional flow-matching equivalence §2.15

$$\mathcal{L}_{\text{CFM}} = \mathcal{L}_{\text{FM}} + \text{const}, \quad \nabla_\theta \mathcal{L}_{\text{CFM}} = \nabla_\theta \mathcal{L}_{\text{FM}}$$

Stochastic dynamics

SDE and Euler–Maruyama step §3.4

$$dX_t = f(X_t, t) dt + g(t) dW_t$$

$$x_{k+1} = x_k + fh + g\sqrt{h}\varepsilon_k$$

Itô's lemma §3.9

$$d\varphi(X_t) = \left[b \cdot \nabla \varphi + \frac{1}{2} \text{tr}(\sigma \sigma^\top \nabla^2 \varphi) \right] dt + \nabla \varphi^\top \sigma dW_t$$

Fokker–Planck equation §3.10

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{1}{2} g^2 \nabla^2 p_t$$

Heat equation §1.32.1

$$\partial_t p_t = D \nabla^2 p_t$$

Ornstein–Uhlenbeck process §3.6

$$dX_t = -\kappa X_t dt + g dW_t$$

$$X_t | x_0 \sim \mathcal{N}\left(e^{-\kappa t} x_0, \frac{g^2}{2\kappa} (1 - e^{-2\kappa t}) I\right)$$

Diffusion

Variance-preserving corruption §3.11

$$X_r = \alpha_r X_0 + \sigma_r \varepsilon, \quad \alpha_r^2 + \sigma_r^2 = 1$$

Score, noise and clean estimate §3.17

$$s = -\frac{\varepsilon_\theta}{\sigma_r}, \quad \hat{x}_0 = \frac{x + \sigma_r^2 s}{\alpha_r}$$

Score matching by parts §3.15

$$J(\psi) = \mathbb{E}_p \left[\frac{1}{2} \|s_\psi\|^2 + \nabla \cdot s_\psi \right] + C$$

Reverse-time SDE §3.23

$$dX_t = [f - g^2 s_t] dt + g dW_t, \quad dt < 0$$

Probability-flow ODE §3.32

$$\frac{dX_t}{dt} = f(X_t, t) - \frac{1}{2} g(t)^2 s_t(X_t)$$

DDPM forward chain §3.24

$$x_k = \sqrt{\bar{\alpha}_k} x_0 + \sqrt{1 - \bar{\alpha}_k} \varepsilon, \quad \bar{\alpha}_k = \prod_{i \leq k} \alpha_i$$

DDPM reverse posterior §3.24

$$\tilde{\mu}_k = \frac{\sqrt{\bar{\alpha}_{k-1}} \beta_k x_0 + \sqrt{\bar{\alpha}_k} (1 - \bar{\alpha}_{k-1}) x_k}{1 - \bar{\alpha}_k}$$

$$\tilde{\beta}_k = \frac{\beta_k (1 - \bar{\alpha}_{k-1})}{1 - \bar{\alpha}_k}$$

Guidance

Conditional score §3.37

$$\nabla \log p(x | c) = \nabla \log p(x) + \nabla \log p(c | x)$$

Guidance scale and tilted target §3.38

$$s_t(x) + \gamma \nabla \log p_t(c | x)$$

$$\iff p_t^\gamma(x) \propto p_t(x) \cdot [p_t(c | x)]^\gamma$$

Reward guidance §3.39

$$s^{(R)} = s + \eta \nabla R, \quad p^{(R)} \propto p e^{\eta R}$$

Where the guidance term enters §3.40

$$\text{SDE drift : } f - g^2 (s + \gamma \nabla \log \ell)$$

$$\text{ODE velocity : } f - \frac{1}{2} g^2 (s + \gamma \nabla \log \ell)$$

Classifier-free guidance §3.43

$$F_{\text{CFG}} = F_\emptyset + w (F_c - F_\emptyset)$$

Multi-objective weighting §3.44

$$\tilde{R}_m = \frac{R_m - \mu_m}{\tau_m}, \quad R_\lambda = \sum_m \lambda_m \tilde{R}_m$$

Information measures

Entropy and differential entropy §1.33

$$H(p) = -\int p \log p, \quad H(p, q) = -\int p \log q$$

KL divergence and its decomposition §1.34

$$D_{\text{KL}}(p \| q) = \mathbb{E}_p \left[\log \frac{p}{q} \right] \geq 0$$

$$H(p, q) = H(p) + D_{\text{KL}}(p \| q)$$